SESSION 1 — MORNING

Time Value of Money

The most fundamental concept in all of finance

Session Roadmap — 2 Hours

This session builds from first principles to full project evaluation. Every concept connects to the next.

01 **What is an Interest Rate?**
3 interpretations • components of r

02 **Future Value & Compounding**
FV formula • compounding frequency • EAR

03 **Present Value & Discounting**
PV formula • timelines • multiple cash flows

04 **NPV & IRR**
Project evaluation • decision rules • worked examples

05 **Annuities & Perpetuities**
Mortgage maths • growing perpetuity • Gordon Growth

Learning Objectives

By the end of this session you will be able to:

- **Interpret an interest rate**

  Describe r as a required return, discount rate, or opportunity cost, and decompose it into its risk components.

- **Calculate FV and PV**

  Apply compounding and discounting formulas for single cash flows, including different compounding frequencies and EAR.

- **Work with multiple cash flows**

  Draw cash flow timelines. Compute the PV and FV of uneven and even cash flow series.

- **Apply NPV and IRR decision rules**

  Evaluate projects using NPV and IRR. Know when they conflict and which to trust.

- **Price annuities & perpetuities**

  Use shortcut formulas for mortgages, lease streams, preferred shares, and the Gordon Growth equity model.

SECTION 01

What Is an Interest Rate?

Three Ways to Read an Interest Rate

If £9,500 today and £10,000 in one year are equivalent in value, what is the interest rate?

$$r = (£10{,}000 - £9{,}500) / £9{,}500 = £500 / £9{,}500 = 5.26\%$$

Required Rate of Return

The minimum return an investor must earn to accept the investment. Below this, they prefer to stay in cash.

Discount Rate

The rate that makes future cash flows equivalent to a present-day value. Used to 'bring money back' in time.

Opportunity Cost

What you give up by choosing one option. Spending £9,500 today means forgoing 5.26% in returns.

Decomposing an Interest Rate

r = Real Risk-Free Rate + Inflation + Default Risk + Liquidity + Maturity Premium

Real Risk-Free Rate	~0.5-1%	Return on a completely risk-free asset in a world with no inflation. Reflects time preference for consumption today vs later.
Inflation Premium	~2-3%	Compensates for expected loss of purchasing power. Nominal rate ≈ Real rate + Inflation (Fisher approximation).
Default Risk Premium	0-5%+	Compensation for the possibility of non-payment. Corporate bonds earn more than government bonds for this reason.
Liquidity Premium	~0.5-1.5%	Compensation for assets that are hard to sell quickly without a price discount. Illiquid bonds carry higher yields.
Maturity Premium	~0.5-2%	Longer-dated bonds are more sensitive to rate changes. Investors demand extra return for tying up money longer.

Example — Building a Corporate Bond Yield

Let's build the required yield for a 5-year corporate bond issued by a mid-size UK company:

Component	Premium Added	Cumulative Yield
Real risk-free rate	0.75%	0.75%
+ Inflation premium (expected 2%/yr)	2.00%	2.75%
+ Default risk premium (BBB-rated)	1.80%	4.55%
+ Liquidity premium (medium-size issuer)	0.50%	5.05%
+ Maturity premium (5-year vs 1-year)	0.45%	5.50%

Required yield = 5.50% — this is the minimum investors demand to hold this bond.

If the bond is issued at 5.50% coupon it will price at par (100). At a higher coupon it prices above par, lower coupon below par. This is covered in the Fixed Income session.

SECTION 02

Future Value & Compounding

Future Value — The Power of Compounding

$$FV = PV \times (1 + r)^n$$

PV
Amount invested today

r
Periodic interest rate

n
Number of periods

FV
Amount at end of period

Key insight — compounding is non-linear. Each year you earn interest on interest, so growth accelerates over time.

Year	Formula	FV at 5%	FV at 10%
1	1,000 × (1.05)^1	£1050.00	£1100.00
2	1,000 × (1.05)^2	£1102.50	£1210.00
3	1,000 × (1.05)^3	£1157.63	£1331.00
5	1,000 × (1.05)^5	£1276.28	£1610.51
10	1,000 × (1.05)^10	£1628.89	£2593.74

Starting amount: £1,000 in all cases

Example 1 — Future Value (Single Cash Flow)

QUESTION

You invest £2,500 today in a savings account paying 4% per year.
How much will you have after 6 years?

Step-by-step solution:

1 **Identify inputs**

PV = £2,500 | r = 4% = 0.04 | n = 6 years

2 **Write the formula**

$FV = PV \times (1 + r)^n = 2{,}500 \times (1.04)^6$

3 **Calculate $(1.04)^6$**

$(1.04)^6 = 1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 \times 1.04 = 1.2653$

The extra £663.22 earned beyond the £600 in simple interest is interest-on-interest (the compounding effect).

4 **Final answer**

Compounding Frequency — Same Rate, Different Results

If the nominal rate is 12% p.a., how much does £1,000 grow to after 1 year under different compounding frequencies?

Formula: FV = PV × (1 + r/m)^(m×n) where m = compounding periods per year

Compounding	Periods/yr (m)	Effective Annual Rate	FV after 1 year (PV=£1,000)
Annual	1	12.000%	£1,120.00
Semi-annual	2	12.360%	£1,123.60
Quarterly	4	12.551%	£1,125.51
Monthly	12	12.683%	£1,126.83
Daily	365	12.747%	£1,127.47
Continuous	∞	12.750%	**£1,127.50**

Continuous compounding (limit): FV = PV × e^{rn} = 1,000 × e^0.12 = £1,127.50

More frequent compounding → slightly higher FV. The difference narrows quickly beyond monthly.

Effective Annual Rate (EAR)

The EAR is the true annual return, accounting for compounding frequency. It enables fair comparison across products.

$$EAR = (1 + r/m)^m - 1$$

EXAMPLE — Which account is better?

Account A: 12% p.a., compounded monthly
Account B: 12.3% p.a., compounded annually

Account A: $EAR = (1 + 0.12/12)^{12} - 1 = (1.01)^{12} - 1 = 12.683\%$
Account B: $EAR = 12.300\%$ (already annual)

→ Account A wins despite the lower stated rate — compounding makes it better.

Always compare products using EAR, not the nominal (stated) rate. Credit cards, mortgages and savings accounts all report different bases.

SECTION 03

Present Value & Discounting

Present Value — Discounting Future Cash Flows

$$PV = FV / (1 + r)^n = FV \times (1 + r)^{-n}$$

Discounting is the reverse of compounding. We ask: what is a future cash flow worth in today's money?

The discount factor $(1+r)^{-n}$ is always ≤ 1 — it shrinks future values. The larger r or n, the smaller the discount factor.

Receive £10,000 in...	Discount rate r	PV Today
1 year	10%	£9,091
1 year	15%	£8,696
3 years	10%	£7,513
5 years	10%	£6,209
10 years	10%	£3,855
10 years	15%	£2,472

→ The same £10,000 is worth very different amounts today depending on timing and risk. This is the entire logic of valuation.

Example 2 — Present Value (Choice Problem)

QUESTION

You are offered two investment payoffs:

Option A: £8,000 today

Option B: £12,500 in 5 years

Your required return is 9% p.a. Which do you prefer?

Solution — find the PV of Option B and compare:

1

Write formula

$PV = FV / (1+r)^n = 12{,}500 / (1.09)^5$

2

Calculate $(1.09)^5$

$(1.09)^5 = 1.5386$

3

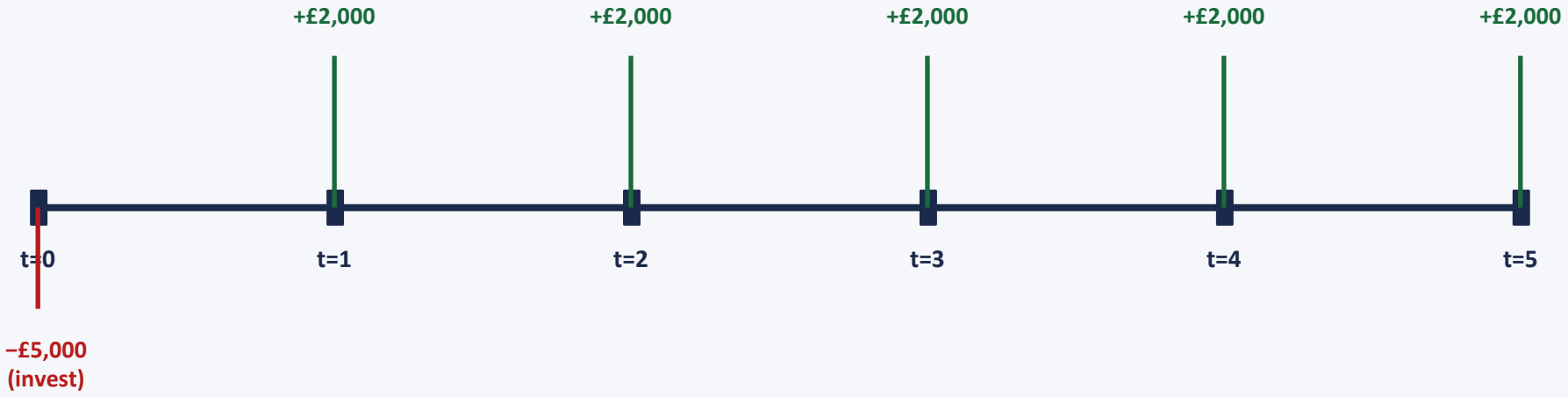
Divide

$PV = 12{,}500 / 1.5386 = £8{,}124.43$

→ Option B is worth £8,124 today vs £8,000 — choose Option B (it has higher PV given your 9% required return).

Cash Flow Timelines — A Critical Tool

Before calculating anything, always draw a timeline. It prevents sign errors and missed cash flows.



Example: Invest £5,000 today, receive £2,000 per year for 5 years. Is this worthwhile at r = 10%?

PV of inflows = $2{,}000/1.1 + 2{,}000/1.1^2 + \ldots + 2{,}000/1.1^5 = 2{,}000 \times [1-(1.1)^{-5}]/0.1$ = £7,581.57

NPV = £7,581.57 − £5,000 = +£2,581.57 → YES, worth doing.

Example 3 — PV of Uneven Cash Flows

QUESTION

A project pays back cash flows as follows: £1,000 at end of year 1, £2,500 at end of year 2, £3,200 at end of year 3, £1,800 at end of year 4.
Your discount rate is 8%. What is the total PV?

Solution — discount each cash flow separately then sum:

Year	Cash Flow	Discount Factor	PV
Year 1	£1,000	$\div (1.08)^1 = \div 1.0800$	£925.93
Year 2	£2,500	$\div (1.08)^2 = \div 1.1664$	£2,143.35
Year 3	£3,200	$\div (1.08)^3 = \div 1.2597$	£2,540.12
Year 4	£1,800	$\div (1.08)^4 = \div 1.3605$	£1,323.06
		TOTAL PV	**£6,932.46**

If the initial investment is less than £6,932 → positive NPV. If more → negative NPV.

SECTION 04

NPV & IRR — Project Evaluation

Net Present Value (NPV)

$$NPV = \Sigma \; CF_t / (1+r)^t - \text{Initial Investment}$$

NPV tells you how much value a project creates in today's money. It is the gold standard of investment appraisal.

NPV > 0	Project generates returns above the discount rate → creates value → ACCEPT
NPV = 0	Project exactly meets the required return → breaks even → indifferent
NPV < 0	Project destroys value → earns less than required return → REJECT

The discount rate r used in NPV should reflect the risk of the project's cash flows — riskier projects use higher r.

Example 4 — Full NPV Calculation

QUESTION

A factory upgrade costs £50,000 today. It is expected to generate cash inflows of £12,000 / £15,000 / £18,000 / £16,000 / £14,000 in years 1–5. Your cost of capital is 10%. Should you invest?

Period	Cash Flow	PV @ 10%
t=0	−£50,000	−£50,000
Year 1	£12,000	£10,909
Year 2	£15,000	£12,397
Year 3	£18,000	£13,524
Year 4	£16,000	£10,928
Year 5	£14,000	£8,693
	Sum of PVs	**£56,451**
	NPV	**£6,451**

✔ ACCEPT

NPV = £6,451

The project earns more than the 10% required return. It creates £6,451 of value.

Internal Rate of Return (IRR)

IRR is the discount rate that makes NPV = 0

Using the same factory example (£50,000 investment, same cash flows):

At r = 10%: NPV = +£4,871 → IRR > 10%

We keep raising the rate until NPV → 0. The IRR is approximately 13.6%.

Discount Rate	NPV
5%	£14,716
10%	£6,451
12%	£3,597
13.6%	**£1,473**
15%	£-279
20%	£-5,824

IRR Decision Rules:

IRR > hurdle rate → Accept project

IRR = hurdle rate → Indifferent

IRR < hurdle rate → Reject project

IRR — Strengths and Limitations

IRR is intuitive and widely used, but has four important limitations that analysts must know.

✔ Strengths

Easy to communicate: "our return is 15%" is intuitive to non-finance people

No discount rate needed: you don't need to pre-specify r to solve for IRR

Useful as a screening tool: quick filter for clearly attractive or unattractive projects

✖ Limitations

Reinvestment assumption: IRR assumes interim cash flows are reinvested at the IRR rate — often unrealistic

Multiple IRRs: if cash flows change sign more than once, multiple IRRs may exist

Scale problem: IRR ignores absolute size — a small project can have high IRR but low NPV

Conflicts with NPV: when projects are mutually exclusive, IRR ranking can differ from NPV ranking

When NPV and IRR conflict, always use NPV. It directly measures value creation in £ terms.

NPV vs IRR — When They Conflict

SCENARIO — Choose one of two mutually exclusive projects (r = 10%):

Project A: Invest £10,000 today → receive £13,500 in 1 year
Project B: Invest £50,000 today → receive £65,000 in 1 year

Project	Investment	Cash Flow yr1	NPV @ 10%	IRR
Project A	£10,000	£13,500	**£2,273**	**35.0%**
Project B	£50,000	£65,000	**£9,091**	**30.0%**

IRR says: choose Project A (35% > 30%)

But Project A's higher return is on a much smaller base. You 'miss' the larger opportunity.

NPV says: choose Project B (+£9,091 vs +£2,273)

NPV measures actual value created in £. Always follow NPV for mutually exclusive decisions.

Rule: NPV always wins for mutually exclusive projects. IRR is fine for standalone accept/reject decisions.

SECTION 05

Annuities & Perpetuities

Ordinary Annuity — Equal Payments at Period End

An annuity is a series of equal payments at equal intervals. Rather than discounting each separately, we use a shortcut:

$$PV = PMT \times [\,1 - (1+r)^{-n}\,] \;/\; r$$

PMT	r	n	[…]
Payment per period	Rate per period	Number of periods	Annuity factor

Applications: mortgage repayments, car loans, bond coupon streams, lease payments, pension payouts.

EXAMPLE — Car Loan

You borrow £20,000 at 6% p.a. (monthly). You will repay in 48 monthly instalments. What is the monthly payment?

r = 6%/12 = 0.5% per month | n = 48 | PV = £20,000

PMT = 20,000 × 0.005 / [1−(1.005)$^{-48}$] = 20,000 × 0.005 / 0.2132 = £469.70/month

Example 5 — Mortgage Cash Flows (CFA Example)

QUESTION (from CFA Level 1 Curriculum)

An investor seeks a 30-year fixed-rate mortgage for 80% of a £1,000,000 property.
The annual mortgage rate is 5.25%. What is the monthly payment?

Step-by-step:

1 **Loan amount**

PV = 80% × £1,000,000 = £800,000

2 **Convert rate**

r = 5.25% / 12 = 0.4375% per month

3 **Convert periods**

n = 30 × 12 = 360 monthly payments

Monthly payment PMT = 800,000 × 0.004375 / [1−(1.004375)$^{-360}$] = £4,417.63

Amortisation — How Payments Split Interest & Principal

Ev e split between interest and principal changes:

Month	Interest Payment	Principal Repayment	Remaining Balance
1	£3,500.00	£917.63	£799,082.37
2	£3,495.99	£921.64	£798,160.73
3	£3,491.95	£925.68	£797,235.05
…	…	…	…
358	£57.48	£4,360.15	£8,777.61
359	£38.40	£4,379.23	£4,398.39
360	£19.24	£4,398.39	£0.00
1	£3,500.00	£917.63	£799,082.37
2	£3,495.99	£921.64	£798,160.73
3	£3,491.95	£925.68	£797,235.05
…	…	…	…
358	£57.48	£4,360.15	£8,777.61
359	£38.40	£4,379.23	£4,398.39
360	£19.24	£4,398.39	£0.00

Key insight: early payments are mostly interest; late payments are mostly principal. Total interest paid over 30 years: £789,347 — nearly equal to the loan itself!

Annuity Due — Payments at Period Beginning

An annuity due shifts every payment one period earlier. Each payment is worth more (one extra period of compounding).

$$PV\ (\text{Annuity Due}) = PV\ (\text{Ordinary Annuity}) \times (1 + r)$$

Ordinary Annuity

Payments at END of each period.
First payment at t=1.
Examples: mortgage, bond coupons, most loans.

Annuity Due

Payments at BEGINNING of each period.
First payment at t=0.
Examples: rent, insurance premiums, leasing contracts.

EXAMPLE

Lease of £500/month, 12 months, 6% annual rate:
Ordinary: PV = $500 \times [1-(1.005)^{-12}]/0.005$ = £5,814.17
Annuity Due: PV = £5,814.17 × 1.005 = £5,843.24 (higher, as payments begin earlier)

Perpetuity — Infinite Equal Payments

As n → ∞ in the annuity formula, the annuity factor converges to 1/r. Simple and elegant:

$$PV = PMT / r$$

The higher the required return r, the lower the price — investors demand more for the same income stream.

EXAMPLE 1 — Preference Share

Shipline PLC pays a fixed dividend of £1.50/year indefinitely.
Shareholders require a 15% return.
PV = 1.50 / 0.15 = £10.00 per share
If required return rises to 20%: PV = 1.50 / 0.20 = £7.50 (share price falls!)

EXAMPLE 2 — Consol Bond (UK Government Historical)

A consol pays £50/year with no maturity. Required return = 4%.
PV = 50 / 0.04 = £1,250
If rates rise to 5%: PV = 50 / 0.05 = £1,000 (−20% price drop on a 1% rate rise!)

Growing Perpetuity — The Gordon Growth Model

If dividends grow at a constant rate g forever, we have a growing perpetuity:

$$PV = D_1 / (r - g) \quad [\text{requires } r > g]$$

This is the Gordon Growth Model (1956) — the foundation of dividend-based equity valuation. It shows that a stock's value depends on three things:

D_1 (Next dividend)	Higher dividend → higher price. Directly proportional.
r (Required return)	Higher risk / higher rates → higher r → lower price.
g (Growth rate)	Faster growth → higher price. Widening the r−g spread matters most.

QUICK EXAMPLE

A company just paid D_0 = £2.00. Dividends grow at 5%/yr. Required return = 12%.

D_1 = 2.00 × 1.05 = £2.10 PV = 2.10 / (0.12 − 0.05) = 2.10 / 0.07 = £30.00 per share

Example 6 — Gordon Growth Model Sensitivity

BASE CASE

Company pays D_1 = £3.00. Required return r = 10%. Growth rate g = 4%.
PV = 3.00 / (0.10 − 0.04) = 3.00 / 0.06 = £50.00 per share

How sensitive is the stock price to changes in r and g?

g \ r	r=8%	r=10%	r=12%	r=14%
g=3%	£60	£43	£33	£27
g=4%	£75	**£50**	£38	£30
g=5%	£100	£60	£43	£33
g=6%	£150	£75	£50	£38

Key takeaways from sensitivity analysis:

1. Model is highly sensitive to the r−g spread. A 1% narrowing in the spread (e.g. r↓ or g↑) dramatically increases the price.
2. Never use g close to r — the formula blows up. Gordon Growth only works for mature, slowly growing firms.

PRACTICE PROBLEMS

Try these before looking at the answers on the next slide.

Q1 You need £25,000 in 4 years for a house deposit. Your savings account pays 3.5% p.a. How much must you deposit today?

Q2 A project costs £100,000 and generates: £30,000 (yr1), £40,000 (yr2), £35,000 (yr3), £25,000 (yr4). At r=8%, should you invest?

Q3 A bond pays £80 coupon per year forever (perpetuity) and requires a 6.4% return. What is the fair price of the bond?

Q4 You take a 5-year loan of £15,000 at 7% annual interest. What is your annual payment? How much total interest do you pay?

Q5 A stock pays D_1=£1.80. Required return is 9%, expected growth is 3%. What is the stock's intrinsic value?

Practice — Answers & Worked Solutions

A1 PV = 25,000 / $(1.035)^4$ = 25,000 / 1.1475 = £21,786

A2 PV of inflows = 27,778 + 34,294 + 27,783 + 18,375 = £108,230 → NPV = +£8,230 → ACCEPT

A3 PV = 80 / 0.064 = £1,250

A4 PMT = 15,000 × 0.07 / $[1-(1.07)^{-5}]$ = £3,660.97. Total paid = 5×3,660.97 = £18,305. Interest = £3,305

A5 PV = 1.80 / (0.09 − 0.03) = 1.80 / 0.06 = £30.00 per share

Common Mistakes — Don't Fall Into These Traps

These errors cost marks in exams and, more importantly, cost money in the real world:

Using the wrong compounding period

If the rate is annual but payments are monthly, convert: r/12, n×12. Never mix annual rates with monthly periods.

Confusing nominal vs EAR

12% monthly compounded ≠ 12% effective. Always convert to EAR before comparing products.

Treating IRR as the project's return

IRR assumes reinvestment at the IRR rate. In practice, cash flows are often reinvested at a lower rate.

Using Gordon Growth when g ≥ r

If growth equals or exceeds the discount rate, the formula returns infinite value. Only use for mature, stable growers.

Forgetting the initial investment in NPV

NPV = PV of inflows − INITIAL INVESTMENT. The t=0 cash flow is a negative number — include it.

Formula Sheet — Session 1 Reference Card

Keep this for your revision. Every formula in this session in one place.

Future Value:	$FV = PV \times (1 + r)^n$
Present Value:	$PV = FV / (1 + r)^n$
Effective Annual Rate:	$EAR = (1 + r/m)^m - 1$
Net Present Value:	$NPV = \Sigma\, CF_t/(1+r)^t - C_0$
Ordinary Annuity (PV):	$PV = PMT \times [1 - (1+r)^{-n}] / r$
Annuity Due (PV):	$PV = PMT \times [1 - (1+r)^{-n}] / r \times (1+r)$
Perpetuity:	$PV = PMT / r$
Gordon Growth Model:	$PV = D_1 / (r - g)$ [requires $r > g$]

Interest Rate Build-Up: r = Real rate + Inflation + Default + Liquidity + Maturity

KEY TAKEAWAYS

Session 1 — Time Value of Money

- An interest rate is simultaneously a required return, discount rate & opportunity cost
- r = Real rate + Inflation + Default + Liquidity + Maturity premiums
- $FV = PV \times (1+r)^n$ — compounding is non-linear; earlier money grows faster
- $PV = FV/(1+r)^n$ — higher risk or longer wait → lower value today
- NPV > 0 = value creation. NPV always beats IRR for mutually exclusive decisions
- Annuities: $PMT \times [1-(1+r)^{-n}]/r$. Perpetuity: PMT/r. Gordon Growth: $D_1/(r-g)$